

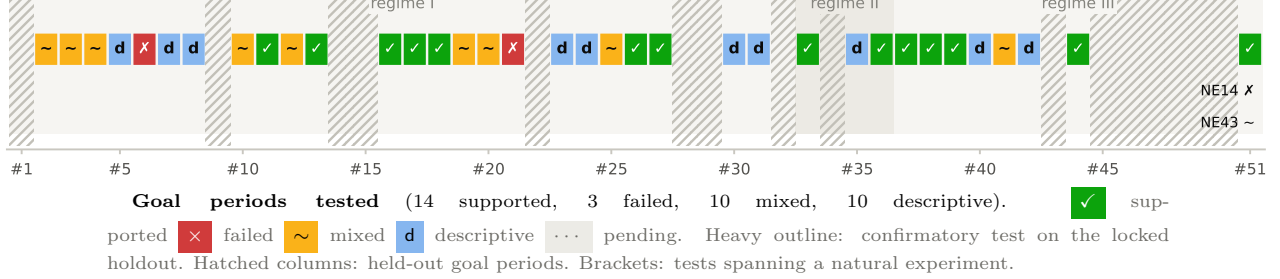
# H86: Taylor's law as a shared-field gauge

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Complete 45% Faithfulness 2/5 Usefulness 2.5/5

**Bottom line.** Taylor's  $c$  measures clock regularity, not the field; the pair-covariance  $c_{\times}$  measures the field, which is small and mostly the scheduler.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



## WHAT HAPPENED

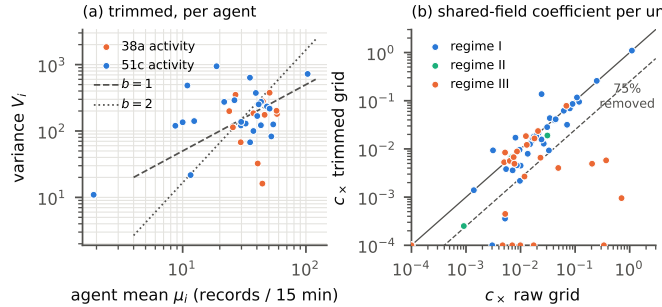
The test asks whether Taylor's law gives a one-number gauge of the shared field. Per period unit, each agent's activity is counted in 15-minute bins, on the raw day window and on DQ8's all-present (trimmed) window. The HH's coefficient  $c_T$  (variance =  $a\mu + c_T\mu^2$  across agents) fails: Taylor's exponent is  $b \approx 0.85$  and  $c_T \leq 0$  in most of 70 units. The pair-covariance coefficient  $c_{\times}$  works as a gauge. It is small (0.016 raw, 0.008 trimmed), and the shared share of excess variance is 0.08. Trimming removes 77% of it in regime III, matching H38's 70–80%. The gauge is now a per-unit column. The holdout script exists and is not run.

## PHYSICAL MODEL

Fluctuation scaling (model 14) with a multiplicative field  $h(t)$  (model 02). Agent  $i$ 's count in bin  $t$  is Poisson with rate  $\mu_i \xi_t \zeta_{it}$ .  $\xi_t$  is a field shared by all present agents (scheduler, kickoff, operator) with variance  $c_{\times}$ .  $\zeta_{it}$  is the agent's private burstiness with variance  $s_i$ .  $c_{\times}$  is read from pair covariances and needs no spread in  $\mu$ .  $\varphi$  is the shared share of super-Poisson variance.

$$V_i = \mu_i + [c_{\times} + s_i(1 + c_{\times})]\mu_i^2, \quad \text{Cov}(y_i, y_j) = c_{\times} \mu_i \mu_j$$

$$c_{\times} = \frac{\sum_{i < j} \text{Cov}_{ij}}{\sum_{i < j} \mu_i \mu_j}, \quad \varphi = \frac{c_{\times} \sum_i \mu_i^2}{\sum_i (V_i - \mu_i)}, \quad r_{\times} = 1 - \frac{c_{\times}^{\text{trim}}}{c_{\times}^{\text{raw}}}$$



**Observables.** (a) Variance against mean per agent, trimmed activity, units 38a and 51c. The points do not follow  $b = 2$ ; faster agents are more regular. (b) Shared-field coefficient per unit, raw against trimmed grid. Regime-III units fall toward the 75%-removed line; regime-I units sit on the diagonal.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Report `taylor_c_shared` on `activity_trim` (and `activity_raw`) from `per_period_estimates` next to any synchrony or influence claim. A trimmed  $c_{\times}$  near 0.008 with  $\varphi < 0.1$  means the swarm's activity fluctuations are about 90% private. Do not use Taylor's  $b$  or  $c_T$  as field measures in a clocked swarm: they track clock regularity. A raw  $c_{\times}$  far above the trimmed one (e.g. 2.2 on 2026-03-31) flags a village-off gap inside the day window. Talk shares about 2.6 times more of its variance than activity ( $\varphi$  0.21 vs 0.08).

## RESULTS VS PREDICTIONS

## FAITHFULNESS SCORECARD

| Prediction                                    | Observed                                    | Verdict        |
|-----------------------------------------------|---------------------------------------------|----------------|
| P1 raw $b \approx 2$ , $c_T > 0$              | $b$ 0.85 [0.21, 1.21]; $c_T > 0$ in 31%     | fail           |
| P2 trimmed $b \leq 1.3$                       | 0.82 (raw already $< 1$ )                   | trivial        |
| P3 III: $r_x \in [0.6, 0.9]$ ; $I < 0.4$      | III 0.77 (13 units); I 0.06 (25)            | pass ( $c_x$ ) |
| P4 kill: III $\varphi_{\text{trim}} \geq 0.5$ | $\varphi$ 0.04; shift null $p < .05$ in 38% | not fired      |
| P5 call clock lowers $b$                      | lower in 58% (need $\geq 67\%$ )            | fail           |
| P6 commits; specificity                       | $b$ 1.26; $\rho$ 0.14 ( $n = 6$ )           | b pass         |
| P7 $r_x$ vs H38 $f_{\text{scaffold}}$         | $\rho$ 0.19 (13 units)                      | fail           |
| P8 talk $\varphi >$ activity                  | 77% of 64 units; 0.21 vs 0.08               | pass           |
| N1 NE14 raw jump                              | one off-gap day; trimmed +0.001             | fail           |
| N2 NE43 operator stops                        | $c_x$ unchanged; $c_T$ no rise              | mixed          |

A mapping 1 F identifiability 2  
 B assumptions 1 G ground truth 1  
 C adequacy 1 H comparative 1  
 D unfitted predictions 1 I transfer 1  
 E interventional 1  
 Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

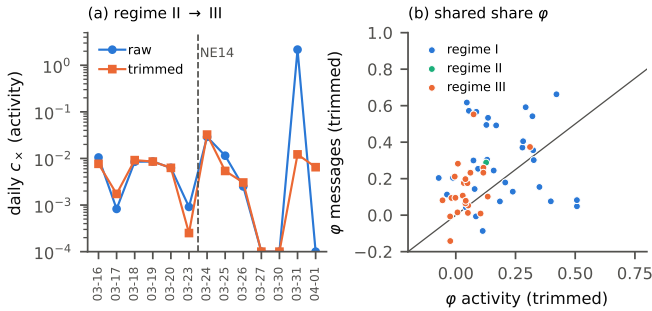
## CAVEATS

Across-agent fits have 4–32 points and a narrow mean range in regime III, so unit-level  $b$  is unstable; the card uses medians. Day-bootstrap CIs under-cover (0.7–0.8 in synthetic runs).  $c_x$  counts equal-time coupling as field. Statistics depend on the 15-minute bin. The commit gauge exists in 13 units, so the kickoff-specificity test has 6 goal periods. The regularity reading of  $b < 1$  is post hoc.

## ROUND 2

Fit a clock-aware Taylor law (renewal model of each agent's calls) and report the residual shared term. Split  $c_x$  into equal-time and one-call-lagged parts on the call clock, which separates field from read-out coupling. Use raw minus trimmed  $c_x$  as a daily off-gap flag. Run the frozen holdout test after sign-off.

- **Essence.** one number per unit that says how much of the swarm's fluctuation is shared, after the scheduler is removed.
- **H86-R1** Clock-aware Taylor law.
- **H86-R2** Lagged vs equal-time  $c_x$ .
- **H86-R3** Off-gap detector.
- **H86-R4** Units 6a and 21b.



**Second figure.** (a) Daily  $c_x$  across NE14 (regime II to III). Raw and trimmed agree except on 2026-03-31, whose window holds a 513-minute all-silent gap. (b) Shared share  $\varphi$  on trimmed grids, messages against activity, one point per unit: talk shares more.

## GOAL PERIODS IN DETAIL

G02 ~ mixed  
 G03 ~ mixed  
 G04 ~ mixed  
 G05 d descriptive  
 G06  $\times$  failed  
 G07 d descriptive  
 G08 d descriptive  
 G10 ~ mixed  
 G11  $\checkmark$  supported  
 G12 ~ mixed  
 +27 more periods: see the card.